

Referee package (DRAFT) – off-anchor STEP-5B closure-margin robustness across the μ^2 band and all intensities

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

This referee document concerns ROBUSTNESS-MU2: whether the STEP-5B layer-closure margin survives OFF the production anchor, across the mass band $\mu^2 \in [\times 0.5, \times 2]$ and all intensities. **Scope:** the closure-margin ratio $K_{\text{budget}}/K(n_{\text{pack}})$ recomputed off-anchor. **Not claimed:** this does not by itself close STEP-5B (the admissible-class exhaustiveness gate is separate); it is robustness evidence.

2. The result

The STEP-5B closure margin ratio = $K_{\text{budget}}/K(n_{\text{pack}})$, recomputed off-anchor, stays > 1 across $\mu^2 \in [\times 0.5, \times 2]$ and all three intensities; worst case $\times 2.55$ at the endpoint, rising to $\sim \times 59$ at production intensity. The margin is therefore not an anchor artefact.

3. Structure of the argument

The Prop-A layer margin and the minimum-transfer envelope are recomputed from a single source (`sectorb_common`) at each μ^2 ; the budget K_{budget} and the consumption $K(n_{\text{pack}})$ are formed and their ratio swept over the band and the intensity grid. Monotonic trends and worst-case values are recorded.

4. Numerical reproduction

Three scripts (margin recompute, band sweep, off-anchor re-margin), each with self-test asserts, reproduce the ratio values and the worst-case $\times 2.55$. Run them from the bundle. The expected minimum ratio is $\times 2.55$ at the band endpoint, and every sampled $(\mu^2, \text{intensity})$ pair satisfies $K_{\text{budget}}/K(n_{\text{pack}}) > 1$; at production intensity the ratio is $\sim \times 59$.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) *Attack the scope.* The result is confined to the scope stated in Sec. 1; a referee may push that boundary, which is fair — the package makes no claim outside it, and the precise scope is in the footer.
- (β) *Attack the tier / over-claim.* No prose here is stronger than the registered grade (footer no-overclaim line); the status is REVIEW DRAFT, NOT operator-confirmed, and the claim-ledger tier is the arbiter.

- (γ) *Attack reproduction / self-containment.* Each reproduction script carries self-test asserts and ships in the folder's bundle with its transitive dependencies (imports and runtime file reads, resolver v1.3.0); a referee runs them and diffs against `expected/`.

Quantitative sanity check (mandatory): the falsifier (footer) is concrete and pre-registered, so the claim is refutable by a single explicit construction; and the numerical values it cites are reproduced by the folder's scripts (all asserts pass).

6. Result footer

Result ID:	B1-RH-ENUM / ROBUSTNESS-MU2 referee package (DRAFT)
Precise statement:	The STEP-5B closure margin $K_{\text{budget}}/K(\text{n_pack})$, recomputed off the production anchor, stays >1 across μ^2 in $[x0.5, x2]$ and all three intensities (worst $x2.55$ endpoint, $\sim x59$ production); the margin is not an anchor artefact.
Scope:	off-anchor STEP-5B closure margin over the μ^2 band. Robustness evidence; STEP-5B gate tracked.
Evidence grade:	robustness EVIDENCE; EXECUTED (3 scripts PASS).
Expected output:	min ratio $x2.55$ (band endpoint), $\sim x59$ (production); all three scripts PASS.
Reproduction command:	python codes/vacuum/robustness_mu2_margin_recompute.py ; robustness_mu2_sweep.py ; robustness_mu2_step5b_remargin.py
Falsifier:	a μ^2 in $[x0.5, x2]$ or an intensity where the recomputed margin ratio ≤ 1 .
Tier before / after:	DRAFT referee package; robustness evidence. DRAFT ACCEPTED (operator 2026-06-11); published-bundle confirmation HELD.
No-overclaim statement:	robustness evidence only; does not close STEP-5B (admissible-class exhaustiveness is separate).
Next required action:	inspect 3-script reproduction bundle -> operator PUBLISHED confirm.